

Khaka - Sketching Companion for Starters

Project Team

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Chapter 1

Introduction

Khaka is an AI-based tool that helps artists and designers start sketching effortlessly by transforming images into detailed sketch layers and guiding them through a step-by-step process to create hand-drawn sketches.

1.1 Problem Statement

Khaka is an application designed to help users create hand-drawn sketches from images. The app breaks down an image into different sketch layers, such as out-lines, shading, and finer details. It provides users with step-by-step instructions on how to manually recreate each layer. As users progress, the app displays the next layer to be sketched, accompanied by text or audio guidance that reads out instructions. This approach helps users follow along and improve their sketching skills, offering clear, structured support throughout the process.

1.2 Scope

Khaka is a web-based application that includes many functionalities for making the life of artists and designers easier. One of the core features of Khaka, which also sets it apart from other apps, is an image-to-layers model that converts the input image into multiple sketch layers, with each layer adding more detail. Before that magic happens, the artistic stylization system allows users to apply modifications to the input image, either through the use of an image-to-image model or manually by adjusting brightness, contrast, saturation, and so on. To further ease the drawing process for the end user, the guidance system, provides instructions at each layer on how to draw that layer. Instructions are presented in textual format; however, if the user prefers, they can be played in audio

format using text-to-speech. If at any stage the user wants feedback on their progress, the Feedback system will help by comparing the drawn sketch with the original one. Once the user has completed their sketch, they can showcase it in their art gallery on Khaka's website for others to view.

1.3 Modules

1.3.1 Image Upload and Processing

- Allow users to upload images from their gallery or take a photo directly within the app.
- Automatically adjust image quality (contrast, brightness) to optimize it for sketch conversion.

1.3.2 Artistic Visualization and Previewing

- Optionally, use an image-to-image model to give the sketch a more artistic look, moving away from realism.
- Users can preview the final result beforehand, ensuring they like what they'll be drawing.
- Convert color images to gray scale for more traditional sketching or allow users to maintain color layers.

1.3.3 Sketch Layer Generation

Using an image as an input create the different sketch layers.

- Outline/Contour Extraction
- Basic Shapes Layers using Loomis or any other well known method
- Shading Layer
- Detail Layer

1.3.4 Layer Display and Navigation

Give the user an ability to view all the layers after their generation and also to go through them.

- **Layer View Toggle:** Enable users to toggle between different layers or view them all together.
- **Layered Progression:** Display the next layer to be sketched as users progress through the steps.
- **Layer Adjustment:** Allow users to adjust the intensity or transparency of each layer.

1.3.5 Step-by-Step Drawing Instructions

Provide step-by-step instructions for each layer, with the option to read the instructions aloud via audio guidance.

- Drawing Instructions using Text
- Drawing Instructions using Audio

1.3.6 Tool recommendation system

- Recommend the types of pencils, pens, or brushes to use for each layer (e.g., fine liner for outlines, soft pencil for shading).

1.3.7 Interactive Feedback

User can upload his drawing and the AI will provide feedback or tips to improve accuracy.

- Take an image of their completed sketch or layer and compare it to the one given by the model

1.3.8 Export and Sharing

Allow the user to share their creation or export the layers and save them some where else.

- **Layer Export:**

- Enable users to export each layer individually.
- Users can also export the end result.
- **Social Sharing:** Integrate options to share sketches on social media.

1.3.9 User Gallery

- A gallery where users can share their sketches and view others' work.

1.4 User Classes and Characteristics

User class	Description
Beginner Artists	Users who are new to drawing or sketching and are seeking guidance to improve their skills.
Intermediate Artists	Users who have some drawing or sketching experience and are looking for tools to refine their skills or try out new techniques.
Professional Artists	Skilled users who are looking for features to help with the more complex sketches that they are trying to create.
Art Educators	Art Educators who teach sketching and create sketching related tutorials might want to use Khaka's functionalities for creating step-by-step tutorials using the layered sketches.
Casual users	Casual users who simply want to see how Khakas functions or just browse and explore other artists' art galleries.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Detailed Use Cases

Following detailed use cases define the actors involved, pre and post conditions how users interact with the system, alternate flow and finally specific requirements.

2.1.1 Manage Art Gallery

Description	Manage the user's personal art gallery by uploading, organizing, editing, and deleting sketches.
Actors	Artist
Preconditions	- User must have an account. - User must be logged in.
Postconditions	User's art gallery is updated.
Main Flow	1. User accesses the gallery. 2. System displays saved sketches. 3. User selects artwork to edit. 4. User makes changes. 5. System saves changes.
Alternate Flow	If no artwork exists, display a message to add new sketches.
Special Requirements	None

Table 2.1: Manage Art Gallery Use Case

2.1.2 Upload Image/Sketch

Description	This use case lets the artist/user upload an image that will be converted into sketching layers.
Actors	Artist
Preconditions	- User must be logged in. - The device must have access to the gallery.
Postconditions	- Image or sketch is uploaded to the application.
Main Flow	1. User clicks on the upload button. 2. User selects an image from their device. 3. User confirms the upload. 4. The application processes the image and forwards it to the layering system.
Alternate Flow	1. If the user cancels the upload process at any point, the system will discard the image and return to the previous state. 2. If the image extension/format is not supported, the system will show an error and prompt the user to choose a different image. 3. If the image size is greater than the allowed limit, the system will display an error message and prompt the user to choose a smaller image.
Special Requirements	- The image uploading process should complete within 3 seconds for files that are under 5 MB. - System must support common image formats such as JPEG, PNG, and BMP. - Uploaded images should be checked for malicious content to ensure application security.

Table 2.2: Upload Image/Sketch Use Case

2.1.3 Apply Artistic Stylization to Image

Description	This use case allows the artist/user to apply artistic styles to the uploaded image, turning it from a realistic depiction to a more artistic one.
Actors	Artist
Preconditions	- User must have uploaded an image. - The user must be logged in.
Postconditions	- The image is converted into an artistic/stylized version, ready for further processing or to be broken down into sketching layers.
Main Flow	1. User navigates to the artistic stylization options. 2. User writes a prompt mentioning the artistic style and gives it to the model. 3. The model processes the selected image and applies the chosen style. 4. The system displays the stylized version of the image to the user.
Alternate Flow	1. If the user clicks the cancel button, the system cancels the changes and returns to the previous image view. 2. If the chosen style fails to apply, the system displays an error message and then prompts the user to try again. 3. If the user wants to change the style after previewing, they can choose a new style. The system will reprocess the image after a changed style.
Special Requirements	- Application should maintain the original quality of the image.

Table 2.3: Apply Artistic Stylization to Image Use Case

2.1.4 Preview the Final Result of Sketch

Description	This use case allows the artist/user to preview the final sketch created from the stylized image before proceeding to the next stage.
Actors	Artist
Preconditions	- The user must have uploaded an image. - The user must be logged in.
Postconditions	- The artist/user has reviewed the final sketch and can either decide to proceed with it or return to make new changes.
Main Flow	1. User clicks the final sketch button from the user interface. 2. System retrieves the stylized image and displays it on the screen.
Alternate Flow	1. If the preview failed due to some error, the system displays an error message and prompts the user to try once again. 2. If the user is not satisfied with the final preview, they can return to the stylization options to make adjustments again.
Special Requirements	None

Table 2.4: Preview the Final Result of Sketch Use Case

2.1.5 Break Down Image into Layers

Description	This use case allows the artist/user to convert the uploaded image into multiple sketch layers of varying details.
Actors	Artist, System
Preconditions	- The user must have uploaded an image. - The user must be satisfied with the final result. - The user must be logged in.
Postconditions	- Image is successfully divided into distinct sketch layers, such as outlines, basic shapes, shading, and detail layers.
Main Flow	1. User clicks on the next button. 2. AI model processes the uploaded image and generates the following layers: • Outline/Contour Layer • Basic Shapes Layer • Shading Layer • Detail Layer 3. System saves the generated layers in user session.
Alternate Flow	1. If the image processing fails due to system error, system displays an error message and prompts the user to try again.
Special Requirements	None

Table 2.5: Break Down Image into Layers Use Case

2.1.6 Toggle Between Different Layers

Description	This use case allows the artist/user to toggle between different sketch layers that are generated in the breakdown stage.
Actors	Artist
Preconditions	- The user must have uploaded an image. - The user must have generated layers for the uploaded image. - The user must be logged in.
Postconditions	- User is able to view and toggle between different sketch layers such as Outline, Basic Shapes, Shading, and Detail layers.
Main Flow	1. System displays a list of all available layers. 2. User can click on "next" or "previous" button to toggle between layers. 3. System shows each layer with its detailed info.
Alternate Flow	1. If no layers are available to switch between, system displays an error message indicating no layers are present for toggling.
Special Requirements	None

Table 2.6: Toggle Between Different Layers Use Case

2.1.7 Get Instructions for Current Layer

Description	This use case allows the artist/user to receive instructions or guidance from AI on how to go about drawing the current layer, based on the level of detail and the overall sketching process.
Actors	Artist, System
Preconditions	- The user must have generated layers for the uploaded image. - The user must be logged in.
Postconditions	- The user receives instructions for the current layer, either in textual or auditory format.
Main Flow	1. User selects a specific sketch layer to work on. 2. System analyzes the content of the selected layer and the context of the sketch. 3. System displays step-by-step instructions, highlighting the areas of focus within the current layer. 4. User follows the provided instructions and can request additional guidance if needed.
Alternate Flow	1. If no specific instructions are available for the selected layer, system notifies the user that instructions are not applicable for the current layer.
Special Requirements	None

Table 2.7: Get Instructions for Current Layer Use Case

2.1.8 Receive Recommendations for Tools

Description	This use case allows the artist/user to receive recommendations for specific tools (e.g. pencils, brushes etc) based on the current sketch layer.
Actors	Artist, System
Preconditions	- The user must have generated layers for the uploaded image. - The user must have a sketch layer selected. - The user must be logged in.
Postconditions	- User gets tool recommendations that are suitable for the current layer such as thin pencil for outlining, soft brush for shading and so on.
Main Flow	1. User selects a specific sketch layer. 2. System analyzes the current sketch layer and the type of sketching activity (e.g. outlining, shading, detailing). 3. System recommends appropriate tools, such as different pencil types, brush sizes, or erasers. 4. User can choose to accept the recommended tools or select their own tools manually.
Alternate Flow	None
Special Requirements	None

Table 2.8: Receive Recommendations for Tools Use Case

2.1.9 Receive Feedback for Drawn Sketch

Description	This use case allows the artist/user to receive feedback on the current sketch, providing suggestions or critiques based on factors such as line quality, proportions, shading accuracy, and overall composition.
Actors	Artist, System
Preconditions	- The user must have completed a sketch layer or reached a certain point where they want feedback. - The user must be logged in.
Postconditions	- System provides feedback on the sketch, highlighting areas for improvement or suggestions for next steps.
Main Flow	1. User completes a layer or section of the sketch and selects the "Feedback" option. 2. System asks the user for an image of the sketch that they have drawn. 3. User uploads the image by pressing the upload button. 4. System analyzes the current sketch using various parameters, such as line consistency, symmetry, and shading techniques. 5. System generates feedback. 6. System displays the feedback to the user, allowing them to view comments and suggested corrections.
Alternate Flow	1. If the system is unable to provide specific feedback, it will display a general message encouraging the user to continue working on the sketch.
Special Requirements	None

Table 2.9: Receive Feedback for Drawn Sketch Use Case

2.1.10 Export/Save Individual Layer

Description	This use case allows the artist/user to export or save an individual sketch layer, enabling them to save a specific stage of their work or share a particular layer with others.
Actors	Artist
Preconditions	- The user must have generated sketch layers. - The user must be logged in.
Postconditions	- The selected sketch layer is saved locally on the user's device.
Main Flow	1. User selects a sketch layer they want to export or save. 2. User clicks on the "Export/Save Layer" option. 3. System asks the user to choose a file format and destination for saving. 4. User selects the format and location and then clicks the confirm button. 5. System saves the layer as a separate image file on the user's device and notifies the user of successful export.
Alternate Flow	1. If the system encounters an error during the export process such as insufficient storage space or permission issues, it displays an error message, asking the user to try again.
Special Requirements	- The application should support multiple file formats for exporting, such as JPEG or PNG for high-quality images.

Table 2.10: Export/Save Individual Layer Use Case

2.1.11 Share Sketch

Description	This use case allows the artist/user to share their sketch with others, either through social media, direct link, or in-app sharing features.
Actors	Artist
Preconditions	- User must be logged in. - Sketch must be saved within the system.
Postconditions	- Selected sketch or layer is successfully shared to the chosen platform.
Main Flow	1. User selects the sketch they would like to share. 2. User clicks on the "Share" button. 3. System prompts the user to choose a sharing option such as social media platform, direct link, or in-app gallery. 4. User selects the option and confirms. 5. System processes the request and generates the appropriate file or link for sharing. 6. System displays a success message, showing that the sketch has been successfully shared.
Alternate Flow	1. If the sharing process fails due to network issues, system displays an error message and prompts the user to try once again.
Special Requirements	None

Table 2.11: Share Sketch Use Case

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is organized by feature as a system feature name and specific functional requirements associated with this feature.

2.2.1 Module 1: Art Gallery Management

1. The system shall allow users to access their personal art gallery from the "Gallery".
2. The system shall display all saved sketches in a grid format.
3. The system shall enable users to select an artwork to edit, update, or delete.

4. If a user has no artwork in their gallery, the system shall display a message indicating that the gallery is empty and provide an option to add new sketches.

2.2.2 Module 2: Image Upload and Processing

1. The system shall allow users to upload images from their devices.
2. The system shall process the uploaded image and forward it to the layering system.
3. If the user cancels the upload, the system shall discard the image and return to the previous state.
4. If the uploaded image format is not supported, the system shall display an error message prompting the user to choose a different image.
5. If the uploaded image exceeds the allowed size limit, the system shall display an error message prompting the user to choose a smaller image.
6. The system shall support common image formats such as JPEG and PNG.

2.2.3 Module 3: Artistic Stylization

1. The system shall allow users to apply artistic styles to the uploaded image.
2. The system shall prompt users to specify the artistic style they wish to apply.
3. The system shall display the stylized version of the image after processing.
4. If the user cancels the stylization process, the system shall revert to the previous image view.
5. If the selected artistic style fails to apply, the system shall display an error message.

2.2.4 Module 4: Sketch Preview

1. The system shall allow users to preview the final sketch created from the stylized image.
2. The system shall retrieve and display the stylized image when the user requests a preview.
3. If the preview fails due to an error, the system shall display an error message.
4. The user shall have the option to return to the stylization options after previewing.

2.2.5 Module 5: Layer Breakdown

1. The system shall convert the uploaded image into multiple sketch layers.
2. The system shall generate the
 - Outline/Contour Layer
 - Basic Shapes Layer
 - Shading Layer
 - Detail Layer
3. The system shall save the generated layers in the user session.
4. If the image processing fails due to a system error, the system shall display an error message.

2.2.6 Module 6: Layer Management

1. The system shall allow users to toggle between different sketch layers.
2. The system shall display a list of all available layers for the user.
3. If no layers are available to switch between, the system shall display an error message.

2.2.7 Module 7: Instruction and Guidance

1. The system shall provide step-by-step instructions for the current layer.
2. The system shall deliver instructions in textual or auditory format based on user preference.
3. If no specific instructions are available, the system shall notify the user.

2.2.8 Module 8: Tool Recommendations

1. The system shall analyze the current sketch layer and recommend specific tools.
2. The user shall have the option to accept the recommended tools or choose their own.

2.2.9 Module 9: Feedback Mechanism

1. The system shall allow users to receive feedback on their current sketch.
2. The user shall upload an image of their sketch for analysis.
3. The system shall analyze the sketch and generate feedback based on various parameters.
4. If the system is unable to provide specific feedback, it shall display a general message encouraging the user to take another image with better light or dark background.

2.2.10 Module 10: Export Functionality

1. The system shall allow users to export or save an individual sketch layer.
2. The user shall select a file format and destination for saving the layer.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are some examples of documenting guidelines.

2.3.1 Performance

App must ensure fast response times to maintain a smooth user experience. Image layer generation should be processed within 60 seconds, even for high-resolution images. File storage should be optimized to minimize load times without compromising quality.

2.3.2 Usability

App's interface should be intuitive and user-friendly, guiding users through the sketching process with clear instructions. Additionally the app should be responsive and functional on all devices.

2.3.3 Security

App should implement strong authentication measures such as two-factor authentication to protect user accounts. Uploaded images and sketches must be handled securely, with encryption for both data in transit and storage. The app should also include mechanisms for content moderation to prevent the sharing of inappropriate or harmful material in the art gallery.

2.3.4 Scalability

As the number of users and images increases, the app must be able to scale efficiently. This requires using cloud storage solutions to handle the growing data volume and implementing caching mechanisms to optimize frequently accessed content. Load balancing should be in place to distribute traffic evenly, ensuring the system performs well even during peak usage.

2.3.5 Reliability

The app should be available for use 24/7, ensuring that users can use it any time. It should be fault tolerant and always ensuring that server issues or high traffic loads aren't affecting the services. Backups should also be created of users uploaded images to protect them from being lost if system fails at any given time.

2.3.6 Maintainability

The app must be easy to maintain, with a modular code. Clear and thorough documentation is also necessary, especially for the APIs and app components.

2.3.7 Compatibility

The app should be compatible so all types of users with any type of browsers can use it freely without any problems. Furthermore, it should be responsive so the site can be viewed on desktop and mobile.

Chapter 3

System Overview

The system will be deployed in a client-server environment, with a tiered architecture ensuring that each functionality is managed independently.

Functionality

This project will be a platform for users where they can upload, process, and interact with images, turning them into multi-layered sketches. The system allows users to apply artistic stylizations to their images, manage individual layers, receive feedback on their work and get instructions on how to recreate the layers.

Design

The design follows a multi-tiered architecture, with three tiers:

- **Presentation Tier:** This includes the user interface and a caching mechanism. The user interface is responsible for all interactions, including image uploads, feedback, and sketch previews. Caching improves performance by storing frequently accessed data.
- **Application Tier:** This is where the business logic is implemented. It consists of several modules:
 - **Authentication:** Handles user logins.
 - **Image Processing:** Applies artistic stylizations and transforms images into layers.
 - **Layering:** Breakdown of images into editable layers.
 - **Feedback and Recommendation Modules:** Feedback and tool suggestions to guide users.
 - **Instruction Module:** Provides textual and auditory instructions to users.

- **Data Tier:** This tier manages all data storage. User data and sketch metadata are stored in a **relational database**, while large image files are kept in a **file storage system**. This ensures efficient storage and fetching of both structured and unstructured data.

3.1 Architectural Design

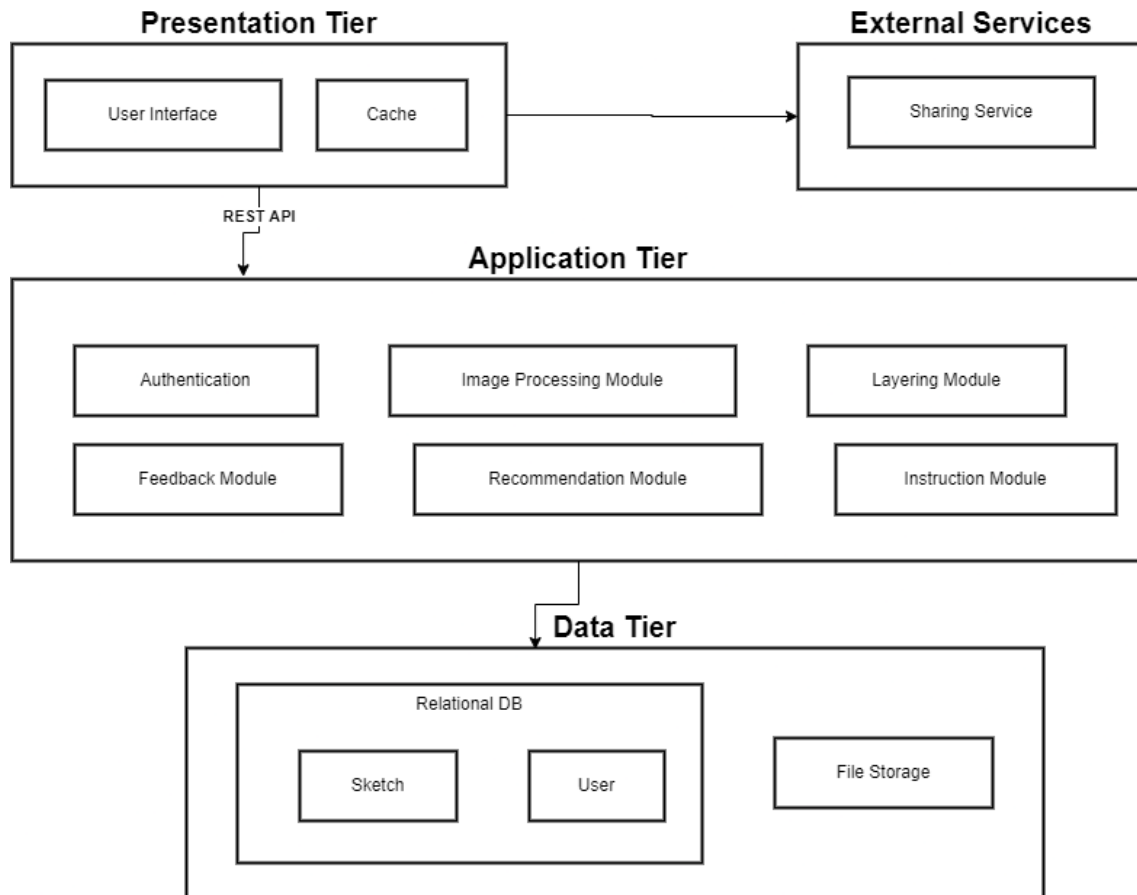


Figure 3.1: Khaka's Architecture Diagram

3.2 Design Models

Design Models for Object Oriented Development Approach

3.2.1 Domain Model

Following domain model diagram shows key entities within the system and their relationships, providing a high-level overview of entire system.

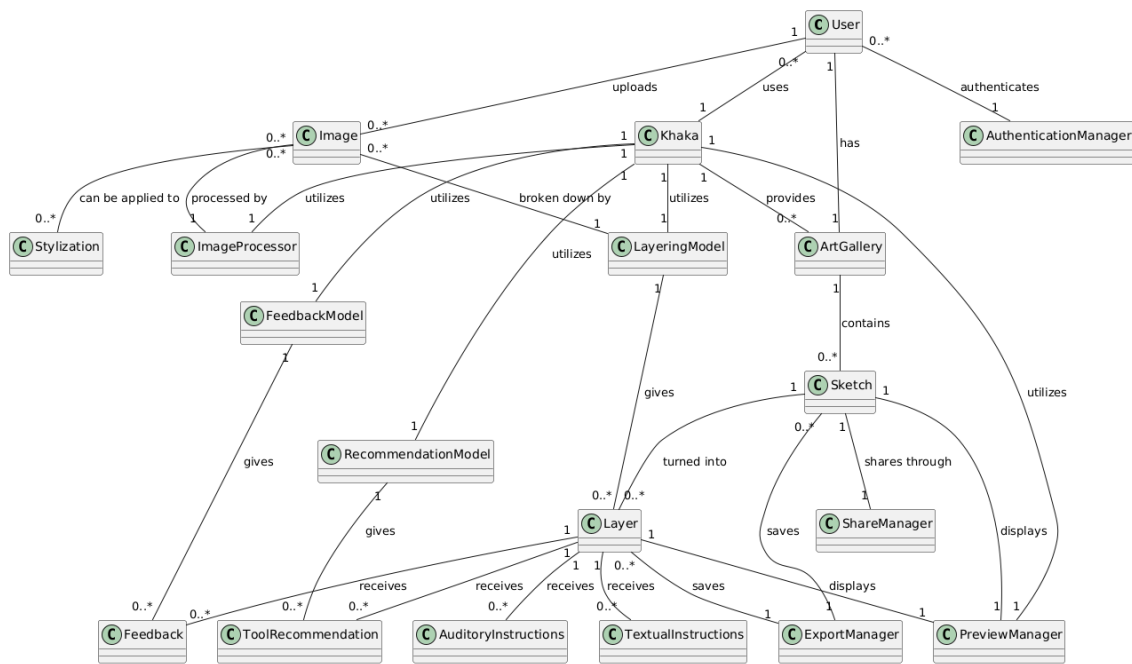


Figure 3.2: Khaka's Domain Model

3.2.2 Activity Diagrams

3.2.2.1 Apply Artistic Stylization

This diagram shows the process of applying artistic styles to user input image and transforming it into a stylized version.

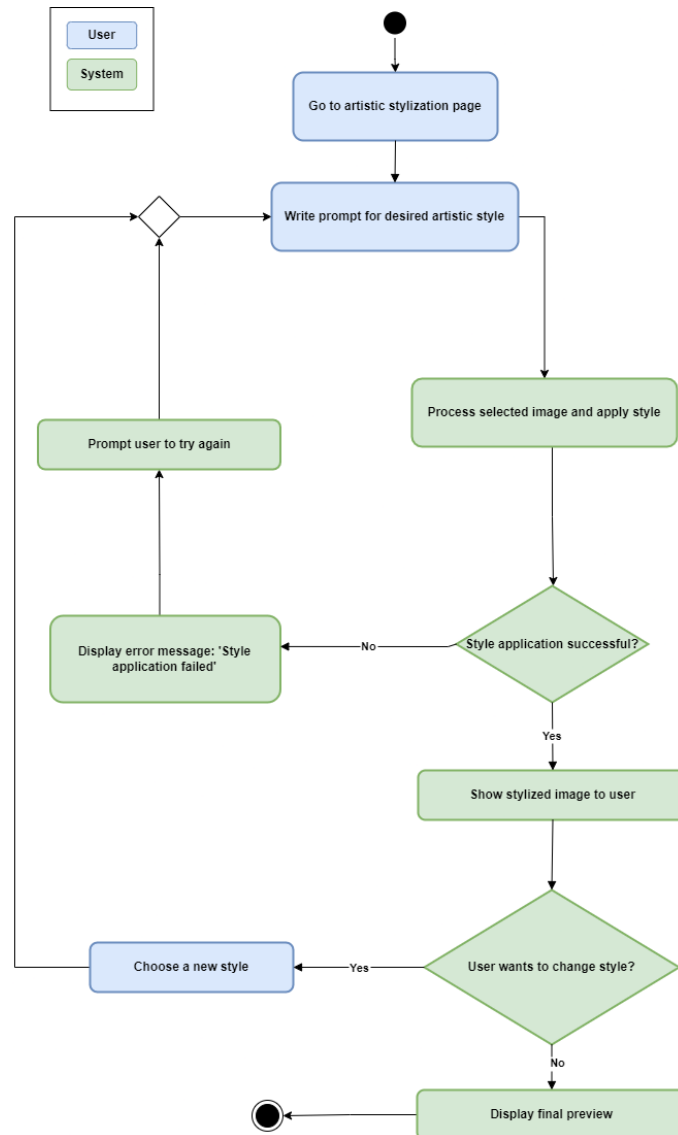


Figure 3.3: Apply Artistic Stylization

3.2.2.2 Breakdown Into Layers

This diagram shows how the uploaded image is divided into multiple layers, each representing different details of the sketch.

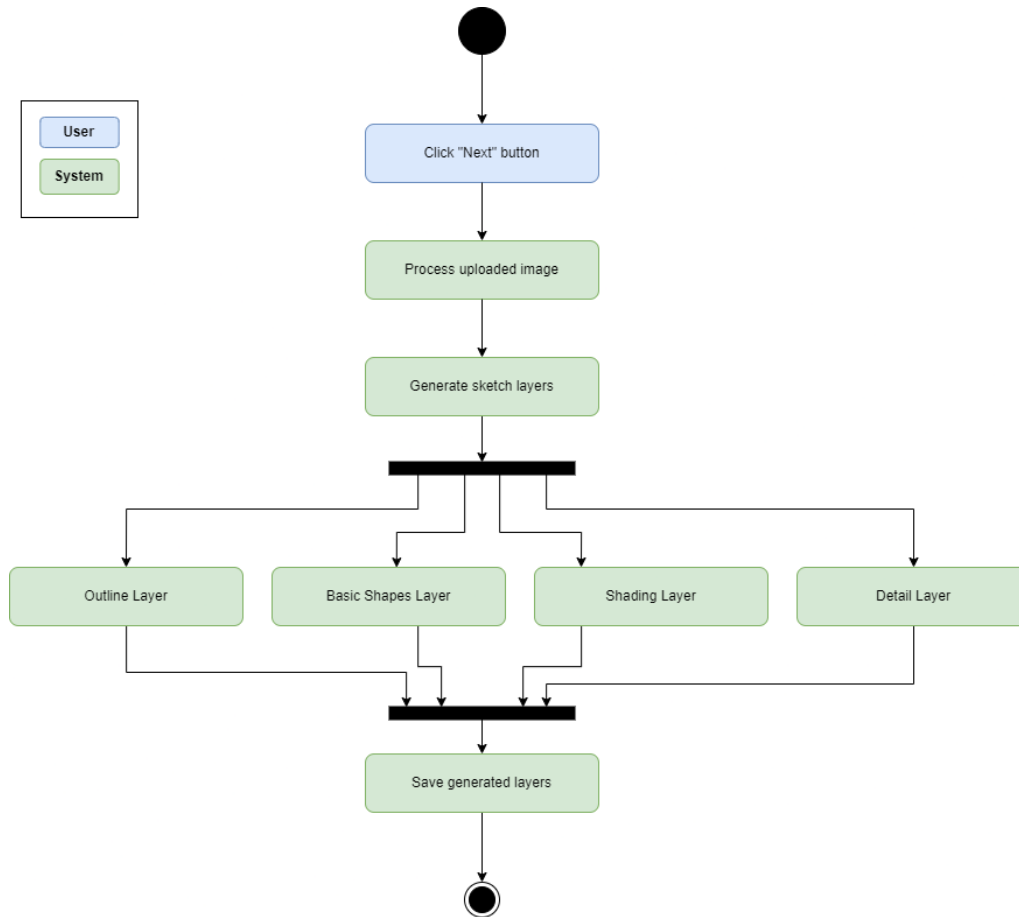


Figure 3.4: Breakdown Into Layers

3.2.2.3 Export Sketch

This diagram outlines the steps required to export the finalized sketch from the application.

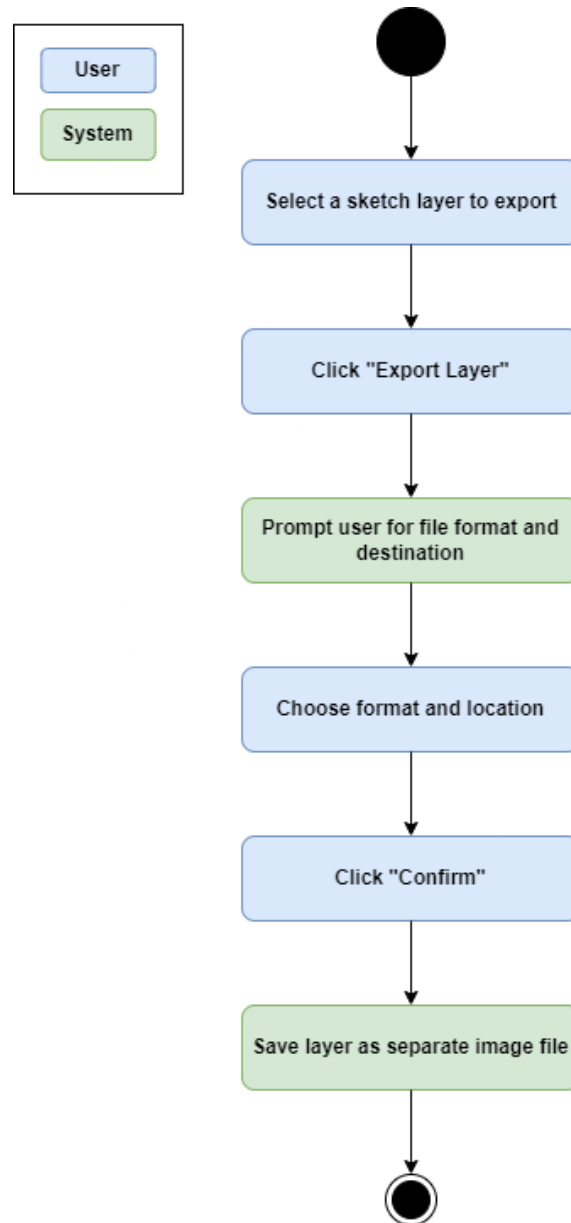


Figure 3.5: Export Sketch

3.2.2.4 Get Feedback

This diagram depicts the process of obtaining feedback from users on the sketches or stylizations applied.

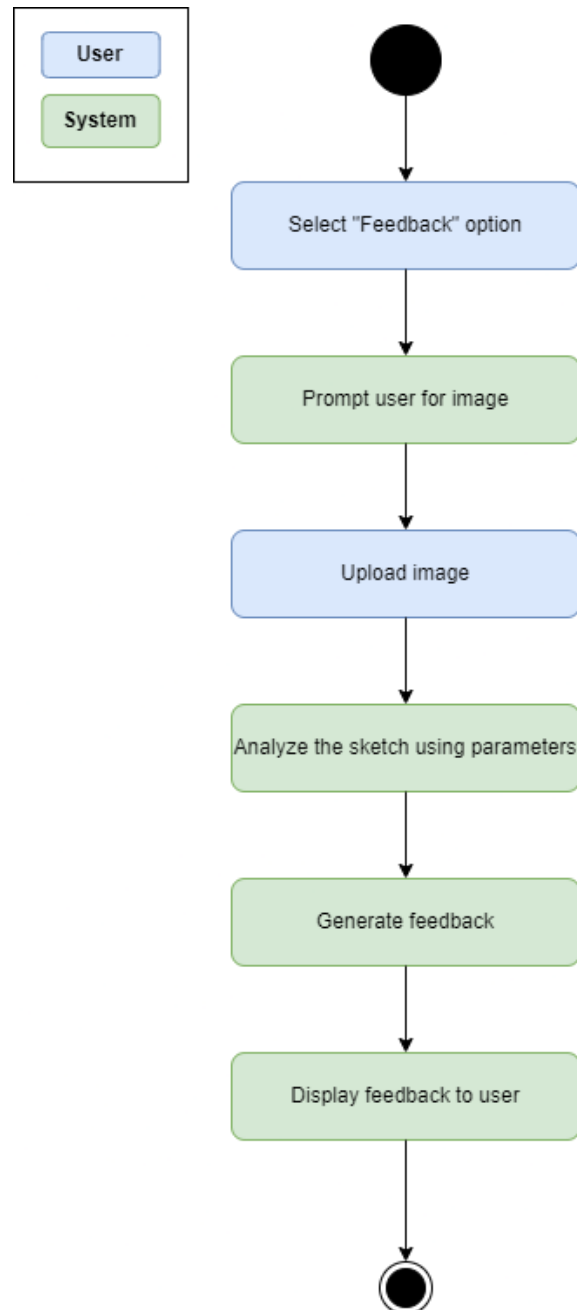


Figure 3.6: Get Feedback

3.2.2.5 Get Instructions

This diagram illustrates how users can access instructions on using various features within the application.

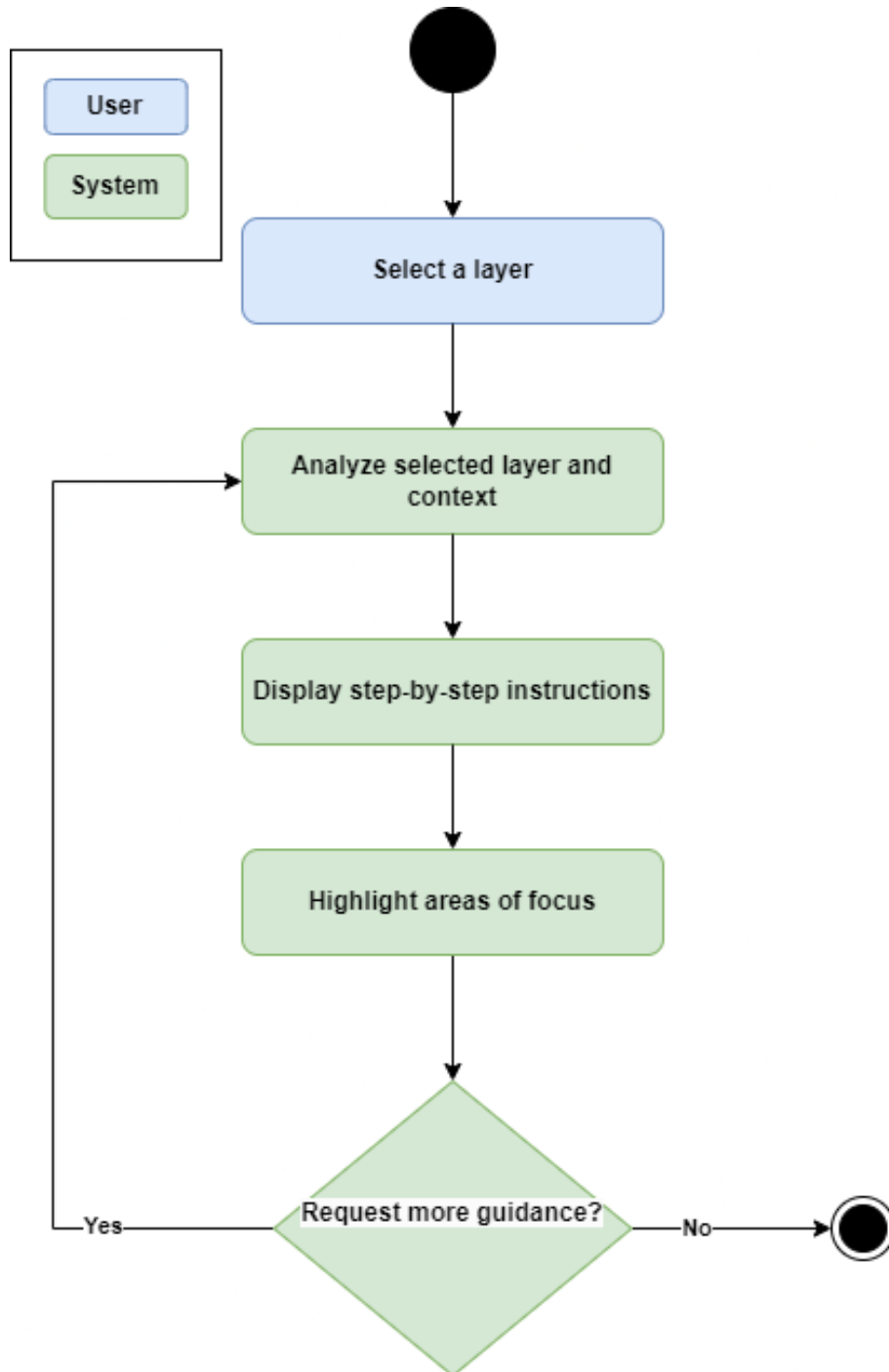


Figure 3.7: Get Instructions

3.2.2.6 Get Recommendations

This diagram shows how the application provides recommendations to users based on their previous activities or preferences.

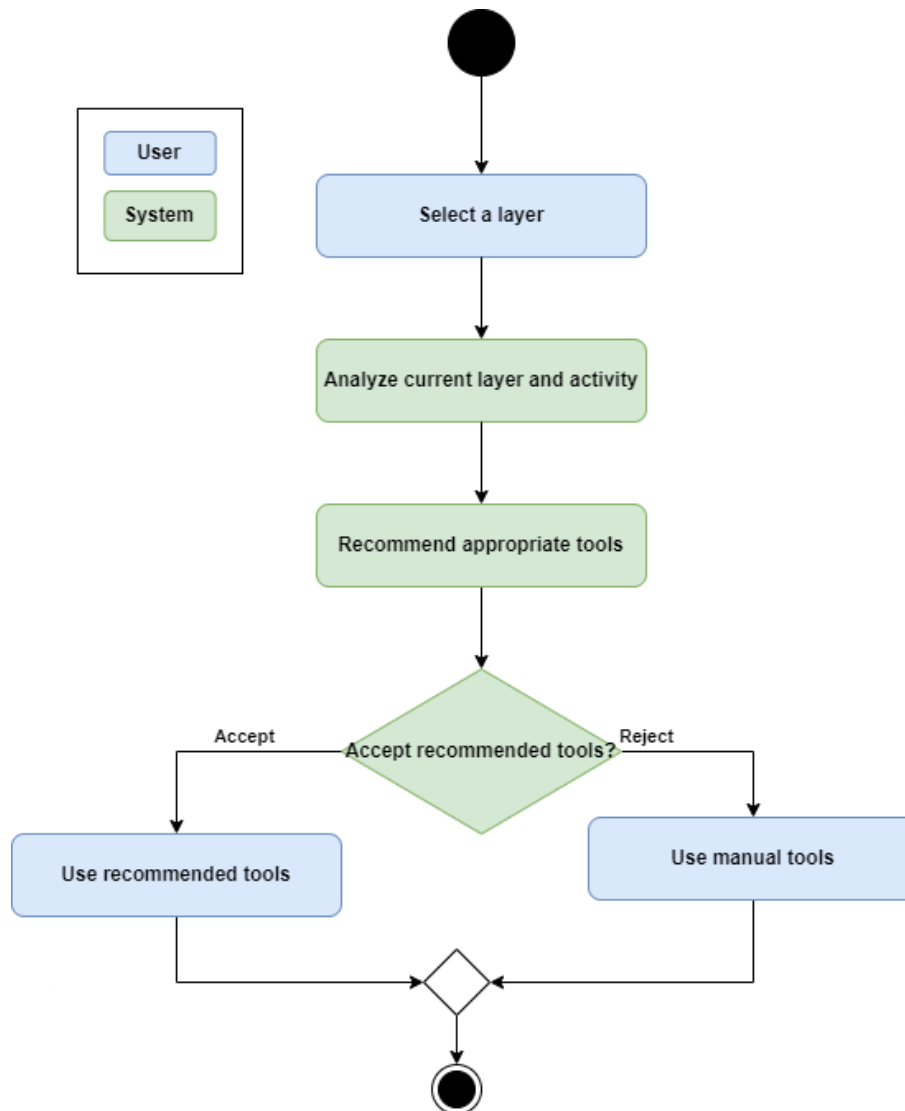


Figure 3.8: Get Recommendations

3.2.2.7 Manage Gallery

This diagram illustrates the steps for managing the artist's personal gallery, including organizing and modifying sketches.

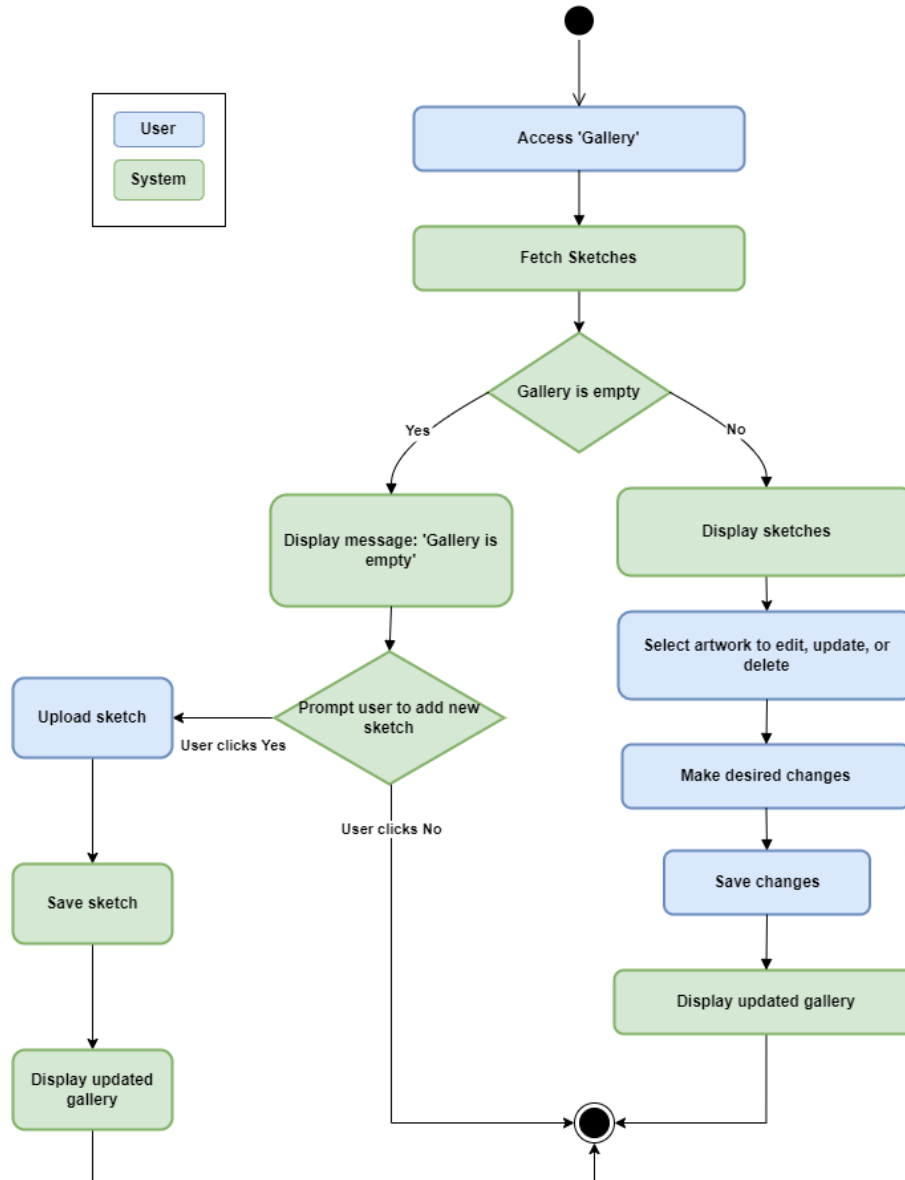


Figure 3.9: Manage Gallery

3.2.2.8 Share Sketch

This diagram outlines the process for sharing sketches with other users or on social media platforms.

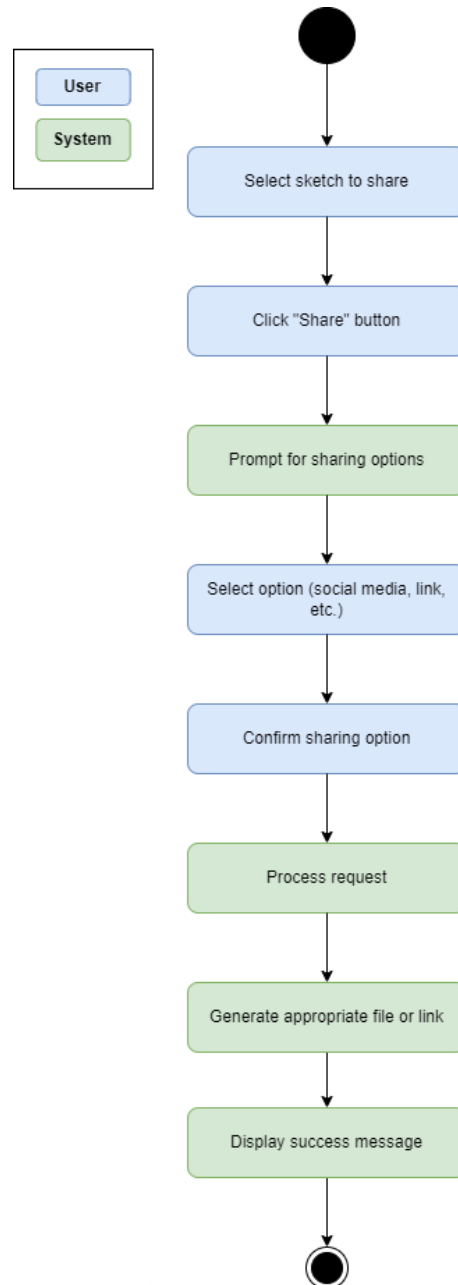


Figure 3.10: Share Sketch

3.2.2.9 Toggle BW Layers

This diagram shows how users can toggle between black and white layers of their sketches for better visibility or processing.

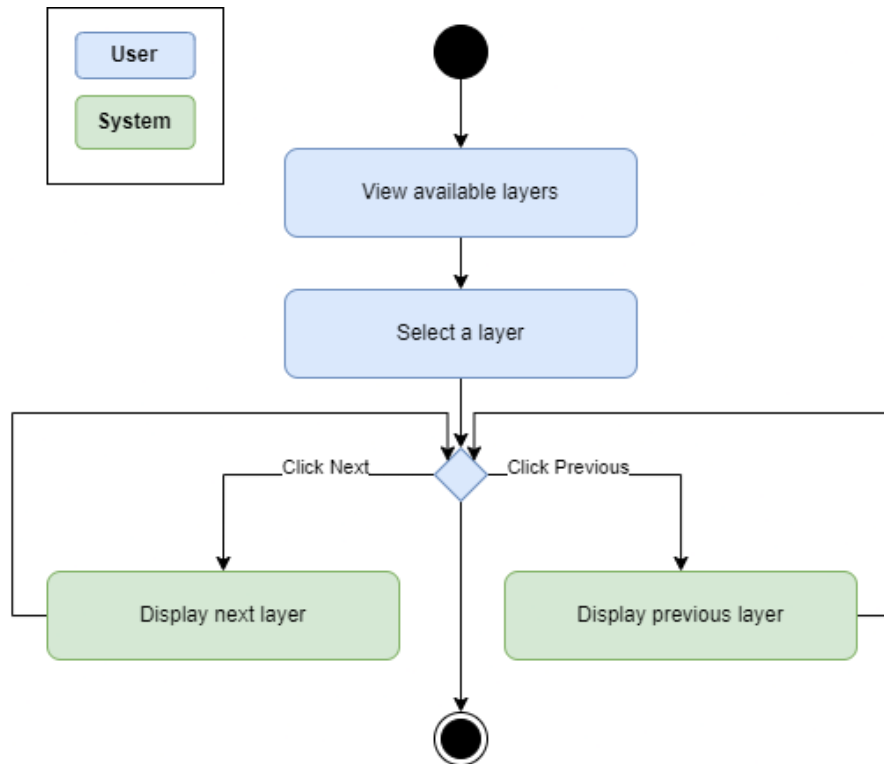


Figure 3.11: Toggle BW Layers

3.2.2.10 Upload Image

This diagram illustrates the steps involved in uploading an image to the application for processing.

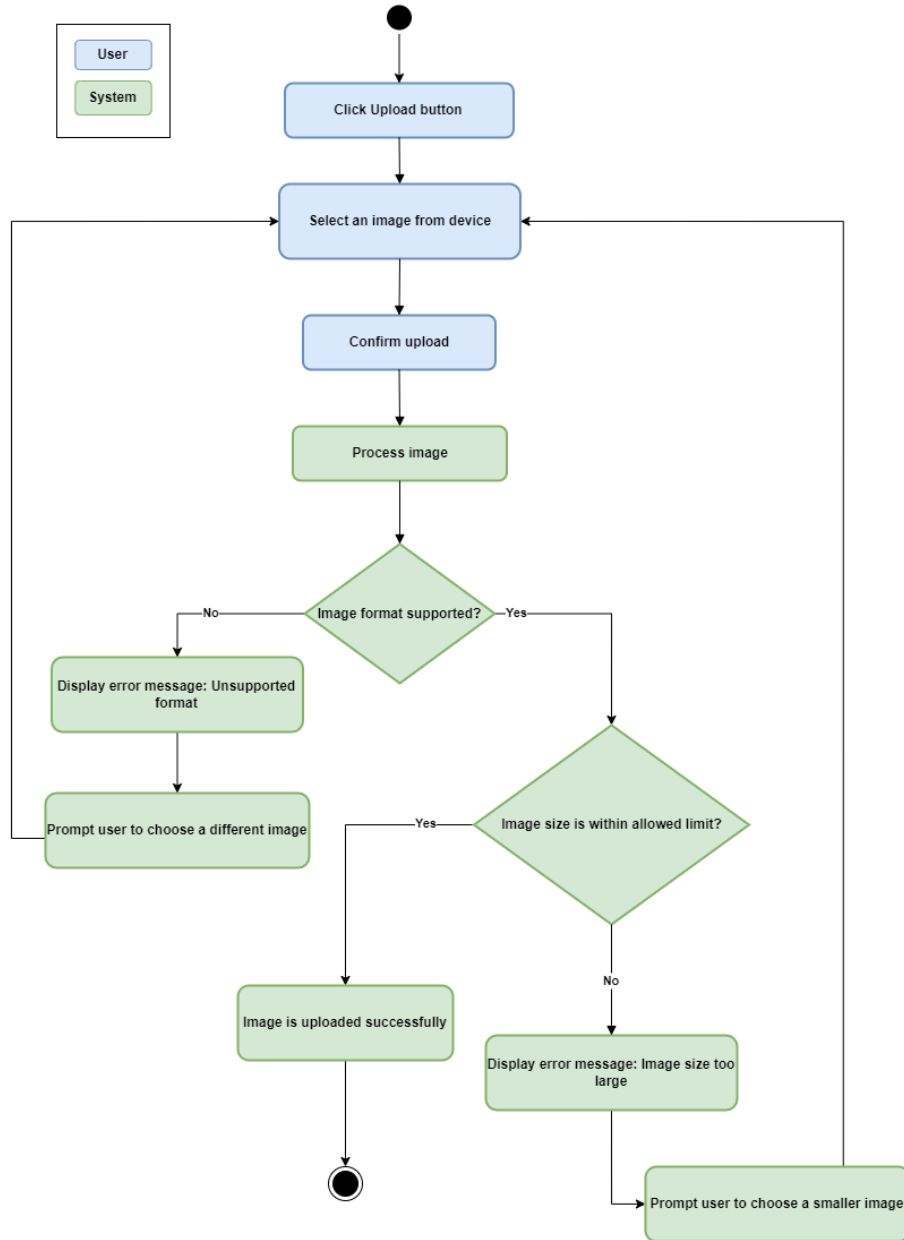


Figure 3.12: Upload Image

3.2.2.11 Overall Activity Diagram

This diagram provides a comprehensive overview of the entire activity flow within the application, summarizing all key processes.

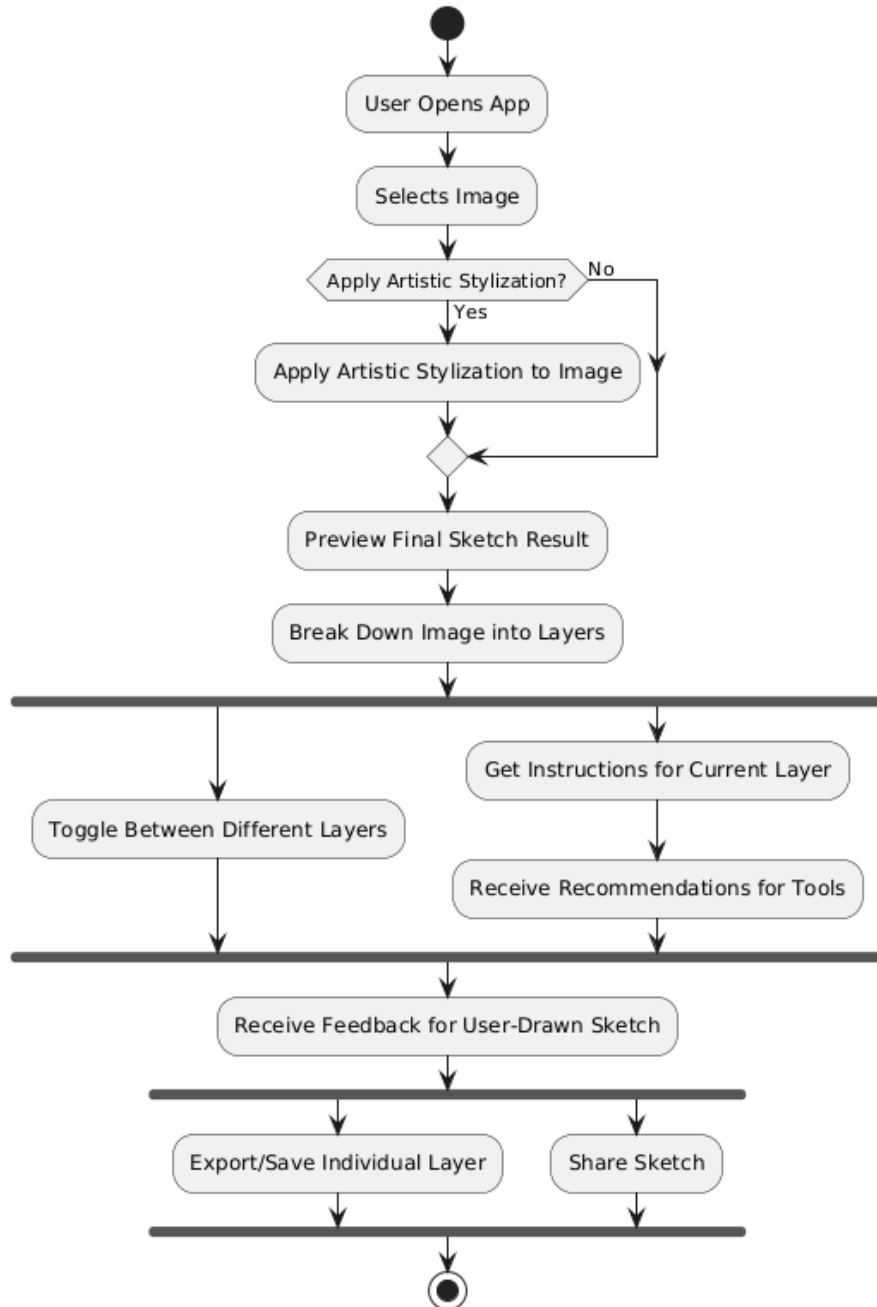


Figure 3.13: Overall Activity Diagram

3.2.3 Class Diagram

The following class diagram shows the main classes in the system, their attributes, methods, and relationships.

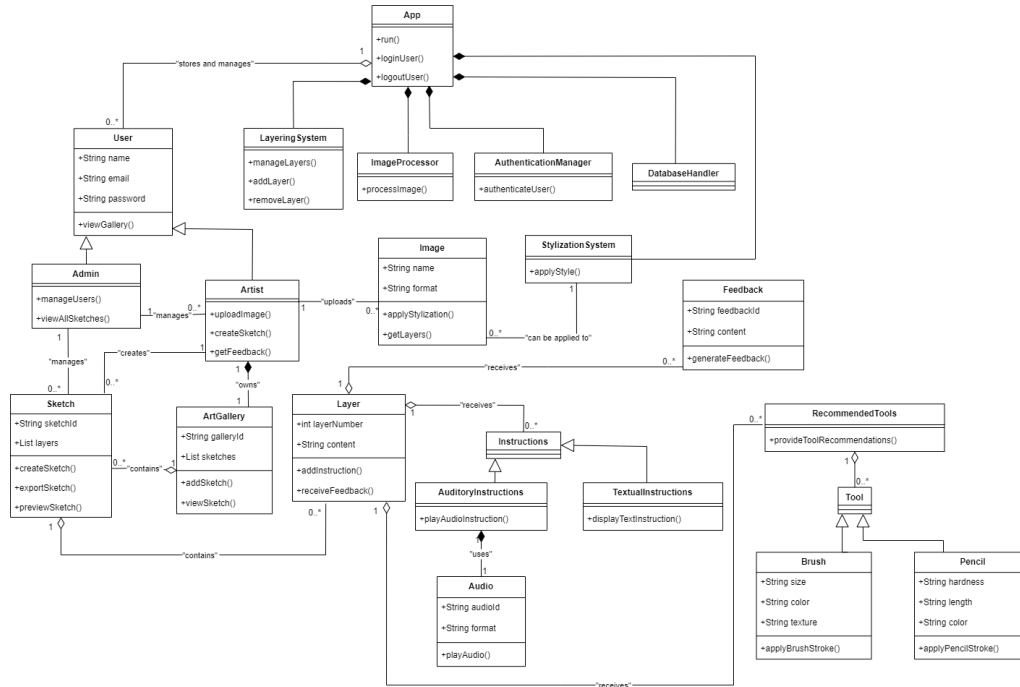


Figure 3.14: Class Diagram

3.2.4 Class-level Sequence Diagram

The following class sequence diagram shows the interactions between different classes over time, illustrating flow of messages and order of operations within the system.

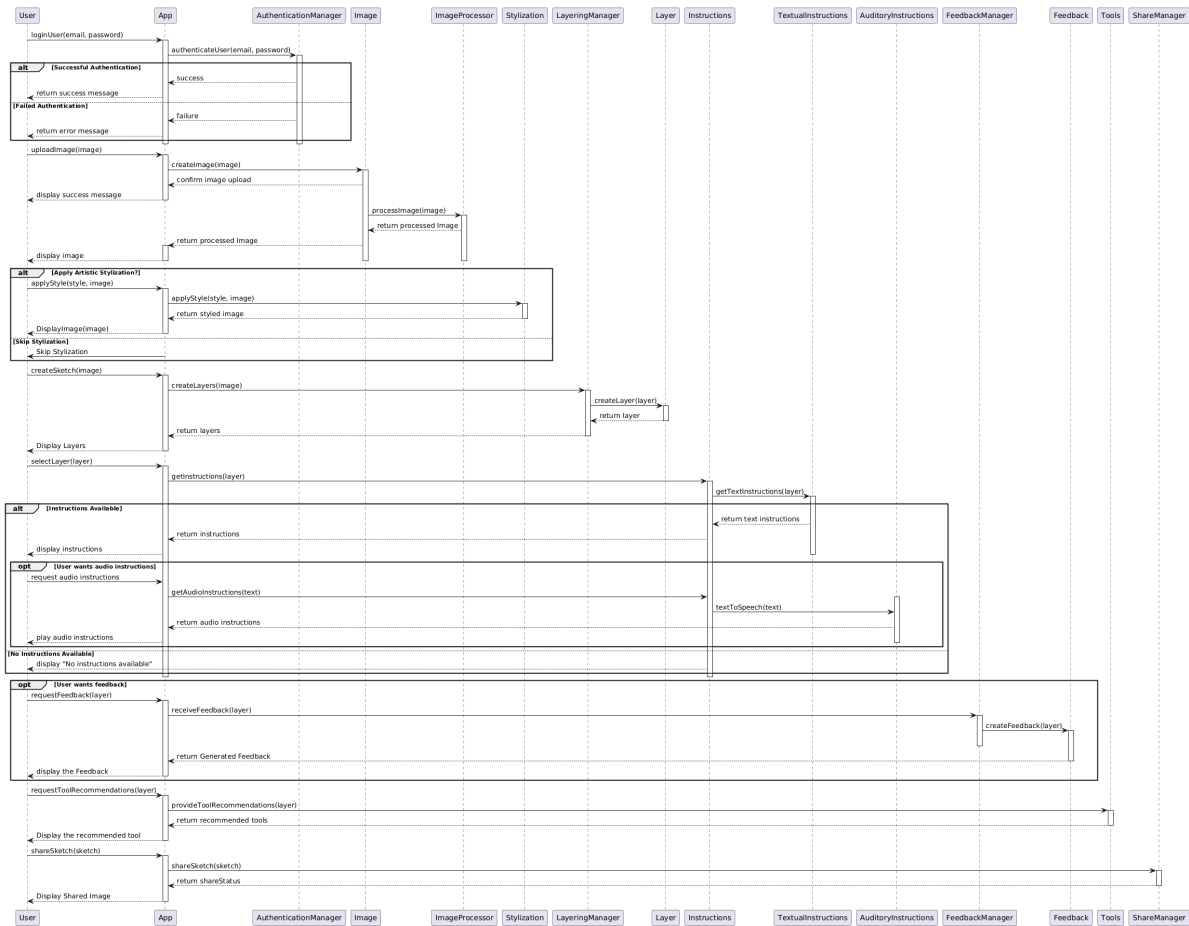


Figure 3.15: Class Level Sequence Diagram

3.2.5 State Transition Diagram

Following state transition diagram depicts various states of a system and also the transitions between these states based on specific events or conditions taking place.

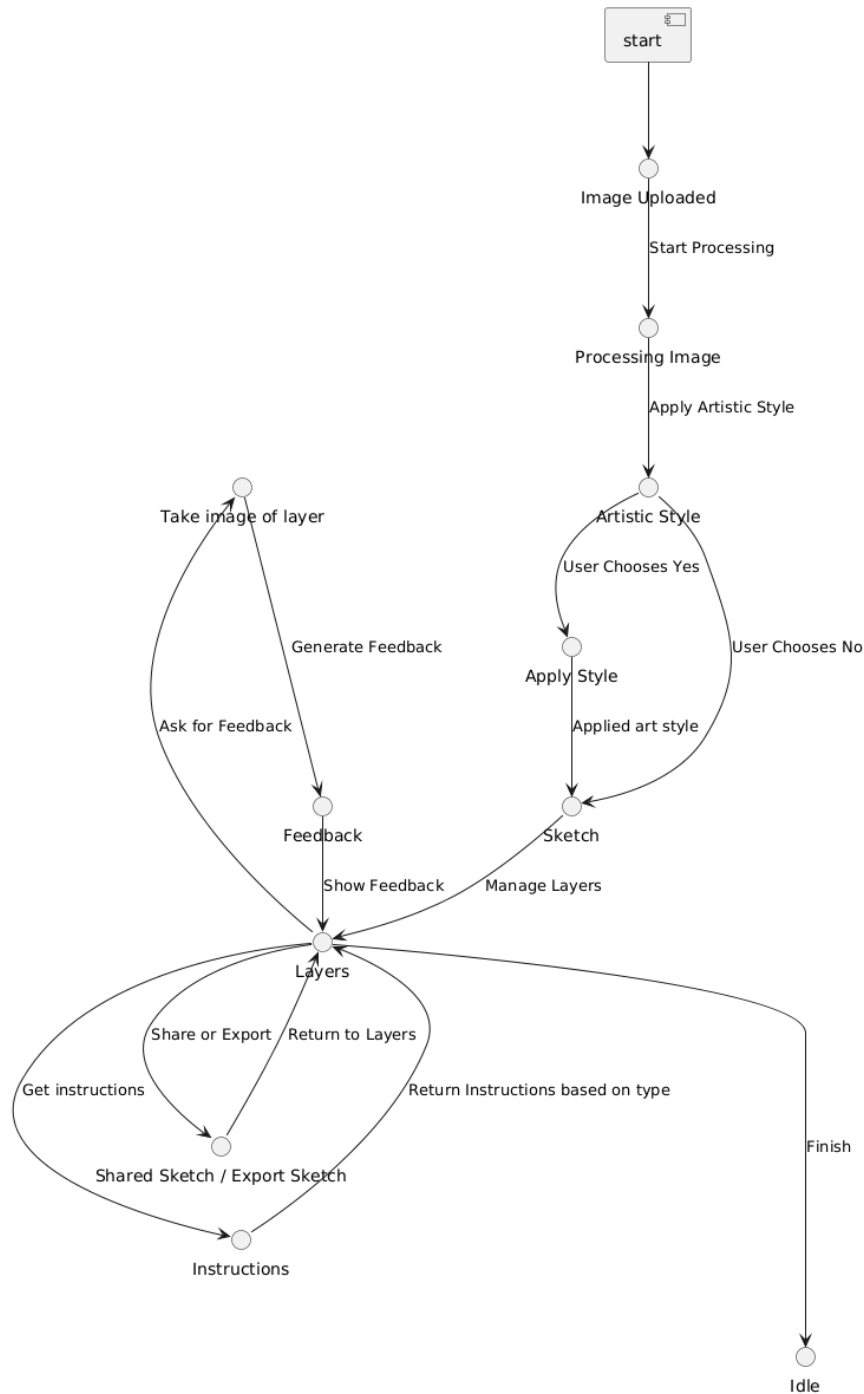


Figure 3.16: State Transition Diagram

3.3 Data Design

The design separates data storage, using MongoDB for user structured data and metadata, while storing actual image and sketch files in a file system storage.

3.3.1 Data Entities and Their Relationships

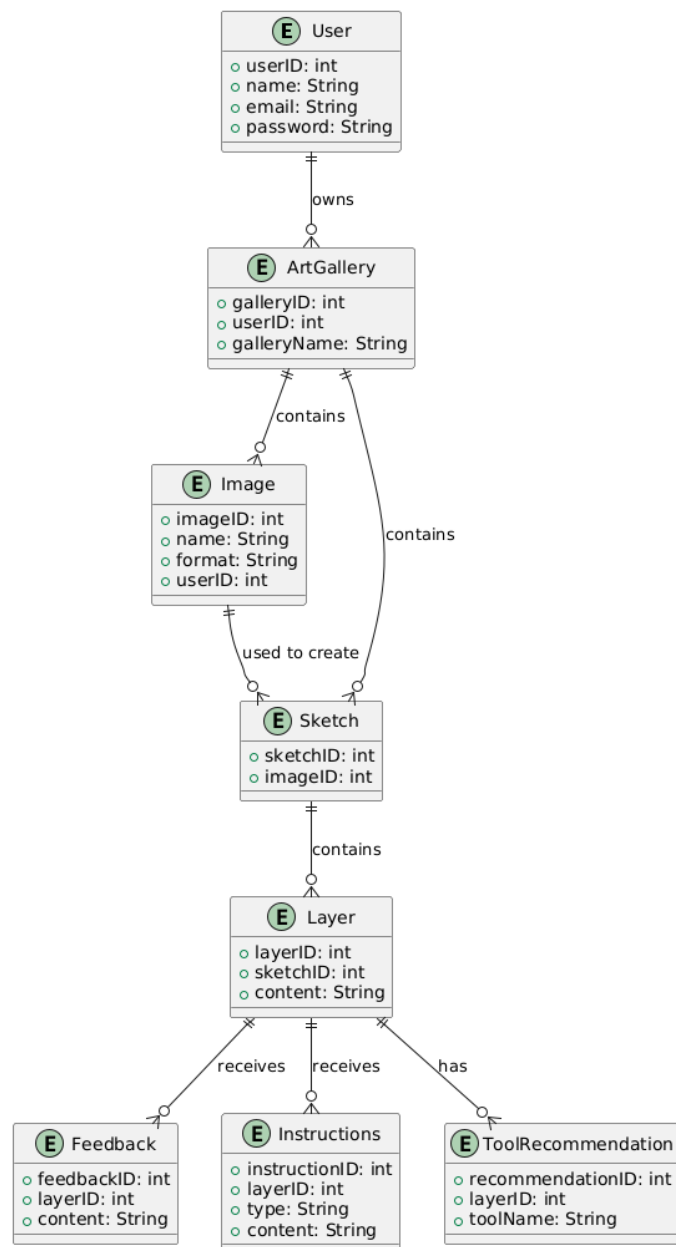


Figure 3.17: Erd Diagram

3.3.2 Databases or Data Storage Items

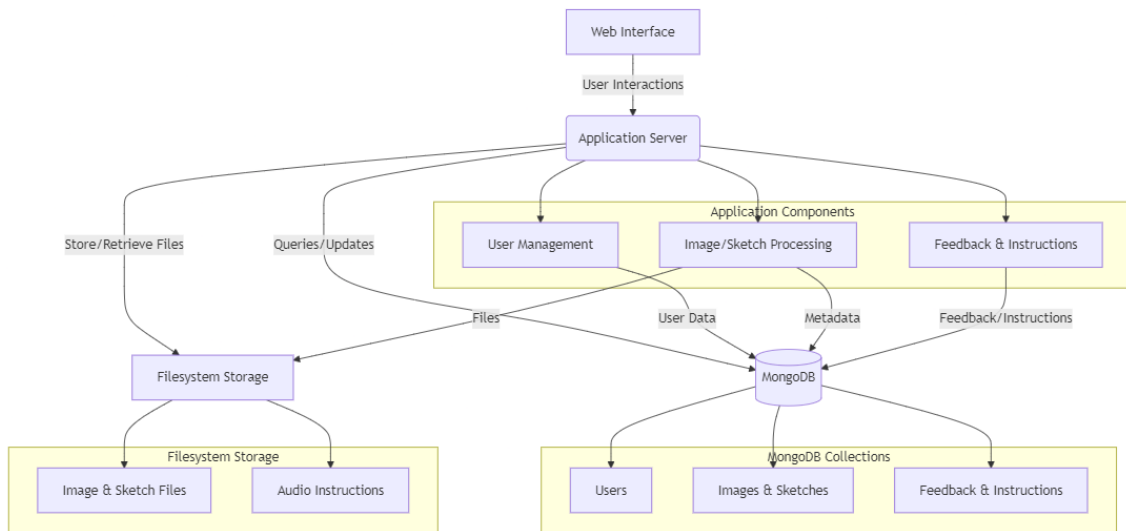


Figure 3.18: Data organization Diagram

Chapter 4

Conclusion

This application is designed with set of functional and non-functional requirements to ensure an amazing experience for users. The project's architecture is structured into three tiers

1. Presentation
2. Application
3. Data

allowing for efficient management of user interactions, image processing, and data storage.

The application has functionalities such as managing art galleries, uploading images, applying artistic styles, generating sketch layers, and providing users with feedback and tool recommendations. All of these features have been explained using use cases.

Furthermore, non-functional requirements have been given for performance, usability, security, scalability, reliability, maintainability, and compatibility.

The integration of all these will create a platform that facilitates artistic expression. As the project progresses these requirements and design will guide in the development and delivering of a polished and effective application.